

Compact polarization splitter-rotator based on lithium niobate-on-insulator platform

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ABSTRACT

We propose a compact mode-coupled polarization splitter-rotator (PSR) based on the lithium niobate-on-insulator platform. The PSR is composed of an asymmetric directional coupler, a taper, and an S-bend. Cross-polarization mode coupling is achieved using a ridge waveguide and air cladding to break the symmetry of the waveguide cross section with a length of only 61 μm . Simulation results show that the insertion losses of this device in the 1500–1600 nm range are less than 1.85 and 0.31 dB for TM_0 and TE_0 modes, respectively, and the extinction ratios are more than 22.5 and 22.7 dB, respectively. In addition, the PSR exhibits a polarization conversion efficiency of 99.97% at 1550 nm. The proposed PSR has great potential for applications in polarization-division multiplexing transmitters and on-chip polarization controllers.

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I. INTRODUCTION

Lithium niobate (LN) has received tremendous attention for its strong electro-optic effect, large nonlinear coefficient, and wide transparency window properties.¹ Conventional methods involve enhancing the refractive index of localized LN crystals through proton exchange or titanium diffusion,^{2,3} thereby forming optical waveguides. However, the refractive index contrast between the core layer of the lumped LN waveguide and the surrounding material is small,⁴ leading to a relatively large mode field and weak nonlinear effect,⁵ which limits its application in photonic integrated circuits (PICs). With the successful preparation of thin-film lithium niobate (TFLN) and the development of micro/nano-processing technology, the development of photonic integrated circuits based on the lithium niobate-on-insulator (LNOI) platform has received great attention. A number of photonic devices have been developed based on such a platform, including wavelength converters,^{6–8} electro-optic modulators,^{9–11} and acousto-optic modulators.^{12–14}

However, the anisotropy of the LN crystal and the high refractive index contrast on the LNOI platform introduce polarization-related dispersion and loss,¹⁵ thereby resulting in a degradation in the performance of integrated devices. A common solution is the

use of polarization-division multiplexing (PDM) technology, where the most important device is the polarization splitter-rotator (PSR). According to the mechanisms, the design of PSRs can be divided into mode evolution and mode coupling. The PSR has been well studied on the silicon-on-insulator (SOI) platform, including various structures: asymmetric directional coupler (ADC) structure,¹⁶ adiabatic taper,¹⁷ inverse-designed structure,¹⁸ subwavelength grating,¹⁹ and discretized subwavelength nanohole structure.²⁰ Currently, the designs of LNOI-based PSRs are based on the principle of mode evolution.^{21–28} Except for Ref. 24, where an extra-long adiabatic taper is used to achieve super mode evolution and directly output TE_0 mode without the TE_1 mode, all other PSRs use adiabatic tapers to achieve the TM_0 mode to evolve into the TE_1 mode after a sufficient length, and then coupling the TE_1 mode into the TE_0 mode. The common feature of the PSRs designed on the LNOI platform is their large size, which is not conducive to on-chip integration.

In this work, we propose a compact PSR based on the principle of mode coupling, which uses both air cladding and ridge waveguide to break the symmetry of the waveguide cross section and achieve direct cross-polarization coupling without a first-order TE mode at the LNOI platform. The overall performance of the device

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is further improved by adding a taper at the output of the waveguide, which increases the spacing between the two waveguides, thereby increasing the converted TE_0 mode. Simulation results show that the proposed PSR has a total length of $61\mu\text{m}$. At the wavelength of 1550 nm , the insertion loss (IL) is less than 0.7 dB , the extinction ratio exceeds 22.5 dB , and the polarization conversion efficiency is 99.97% . In the wavelength range of $1500\text{--}1600\text{ nm}$, the insertion losses are less than 1.85 and 0.31 dB , and the extinction ratios (ERs) exceed 22.5 and 22.7 dB for the TM_0 and TE_0 modes, respectively.

II. STRUCTURE DESIGN AND SIMULATION

The schematic diagram of the proposed PSR is shown in Fig. 1. The proposed PSR consists of two waveguides, WG1 and WG2. WG1 includes a straight waveguide and a linear taper, and WG2 includes a straight waveguide and an S-bend. We perform the device design on an LNOI platform with an LN thickness of 500 nm . The ridge waveguide is etched at a depth H_e of 300 nm , and the waveguide sidewall angle θ is 80° in both WG1 and WG2.²⁹ The propagation direction of the waveguide is chosen to be the crystallographic Z direction (z axis). It is known that the x axis and the y axis correspond with the ordinary refractive index $n_o = 2.211$, and the z axis corresponds with the extraordinary refractive index $n_e = 2.138$ for the wavelength of 1550 nm .³⁰ For the S-bend, its horizontal length and vertical length are $L_x = 1\mu\text{m}$ and $L_z = 20\mu\text{m}$, respectively.

According to the coupled mode theory,³¹ the phase matching between the waveguides must be satisfied to achieve the coupling between two orthogonal modes. This means that the effective refractive index of the TM_0 mode in the coupling region of WG1 should be equal to the effective refractive index of the TE_0 mode in the coupling region of WG2. In addition, the symmetry of the waveguide cross section needs to be broken. The proposed PSR uses both air cladding and ridge waveguide to break the symmetry and achieve direct coupling across polarization.³² To determine the two waveguide widths w_1 and w_2 during phase matching, the effective refractive indices of the first three modes in the LN ridge waveguide are calculated as the waveguide width varies from 0.4 to $1.3\mu\text{m}$, as shown in Fig. 2. From the trade-off between the coupling length and the limitation of the mode field in the waveguide, we

choose $w_1 = 1\mu\text{m}$ and $w_2 = 0.55\mu\text{m}$. The spacing g between the two waveguides is set to 300 nm . The TM_0 mode satisfies the condition of phase matching, thus coupling to become the TE_0 mode in WG2. At the same time, the TE_0 mode exhibits a huge effective refractive index difference between WG1 and WG2, leading to a phase mismatch, and thus, no coupling occurring.

To achieve a more efficient mode coupling, finding the coupling length is an important step. When light is input from the waveguide, it excites two super modes which are the even mode and the odd mode in the waveguide. After a coupling length, the light is coupled from one waveguide to the other. The theoretical coupling length L_c can be calculated as³³

$$L_c = \frac{\lambda}{2(n_{\text{even}} - n_{\text{odd}})}, \quad (1)$$

where n_{even} and n_{odd} are the refractive indices of the even mode and the odd mode, respectively, and λ is the operating wavelength. The theoretical coupling length of the device is calculated to be $46\mu\text{m}$ for the selected waveguide widths and spacing. The transmissions of the TM_0 and TE_0 modes in the waveguide are verified by the three-dimensional finite-difference time-domain (3D-FDTD) method.³⁴ We use perfectly matched layers (PMLs) as boundary conditions. Figure 3(a) shows a plot of the polarization conversion efficiency at the cross and through ends with the coupling length L_c for TM_0 mode incidence. For polarization splitter-rotators, the polarization conversion efficiency is an important parameter to measure the performance of the device, which is denoted as³⁵

$$PCE = 10\lg(T_{\text{cro_TE}}/(T_{\text{cro_TE}} + T_{\text{thr_TM}})), \quad (2)$$

where $T_{\text{cro_TE}}$ and $T_{\text{thr_TM}}$ denote the corresponding transmissions at TM_0 mode incidence. In Fig. 3(a), the polarization coupling efficiency of the device is calculated for a coupling length in the range of $35\text{--}44\mu\text{m}$. The polarization coupling efficiency of the device increases and then decreases, and it is the highest at a coupling length of $41\mu\text{m}$; so we choose a coupling length of $41\mu\text{m}$. The $5\mu\text{m}$ offset from the theoretical coupling length is due to the fact that the S-bend added at the end of the coupling region does not

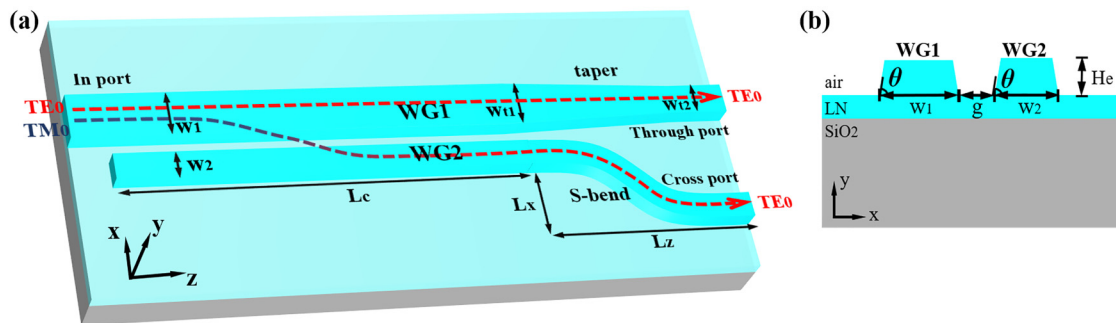


FIG. 1. Schematic structure of the proposed PSR. (a) Top view and (b) cross-sectional view of the device in coupling.

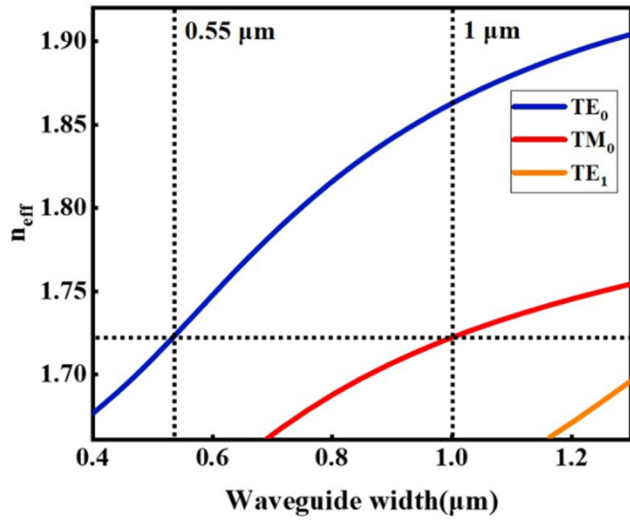


FIG. 2. The calculated effective refractive indices n_{eff} for the first three modes of the ridge waveguide with air cladding as the waveguide width variation.

vary much from the distance of WG2 at the beginning, resulting in some of the light continuing to be coupled. Figures 3(b) and 3(c) show the optical field diagrams at the incidence of the TM_0 and TE_0 modes, respectively, when the coupling length is $41\text{ }\mu\text{m}$. Meanwhile, the theoretical coupling length of the phase-matched coupling to the narrow waveguide is $125\text{ }\mu\text{m}$ along the Y-propagation direction, while the width of the wide waveguide is equal to $1\text{ }\mu\text{m}$. Therefore, we select the Z direction to reduce the coupling length.

The spacing between WG1 and WG2 after the coupling region is not large enough, which may result in partial re-coupling.

We add a taper after the WG1 coupling region to increase the spacing between the two waveguides. The initial width of the taper w_{t1} is equal to w_1 . To determine the optimal length of the taper, Figs. 4(a) and 4(b) show the transmissions of the TM_0 and TE_0 modes for taper lengths of $13\text{--}22\text{ }\mu\text{m}$ when the width of the taper end w_{t2} is set to $0.7\text{ }\mu\text{m}$. The subscripts inside $T_{\text{cro_TE}}$ and $T_{\text{thr_TE}}$ denote the transmissions of the TE_0 mode at the cross-port and the through-port after the incident light has been transmitted in the device, respectively. Similarly, the transmissions of the TM_0 mode at the output ports can be represented as $T_{\text{cro_TM}}$ and $T_{\text{thr_TM}}$. It can be seen that as the taper length increases, the desired cross-port TE_0 mode optical transmission for TM_0 mode incidence increases, and the other undesired modes decrease and reach a minimum at a taper length of $17\text{ }\mu\text{m}$. For TE_0 mode incidence, the extinction ratio is highest at a coupling length of $17\text{ }\mu\text{m}$. Combining the transmissions of the TM_0 and TE_0 modes, the length of the taper is determined to be $17\text{ }\mu\text{m}$. Under the condition of determining the optimal taper length, Figs. 4(c) and 4(d) demonstrate the transmissions of the TM_0 mode and the TE_0 mode in the device when the width of the taper end ranges from 0.5 to $1\text{ }\mu\text{m}$. When the width of the taper end becomes smaller, Fig. 4(c) shows that the TE_0 mode at the cross-port increases and the very few unconverted TM_0 modes at the through-port become even fewer. Combining the transmissions of the TM_0 and TE_0 modes with different taper widths, the width of the taper end is finally selected to be $0.8\text{ }\mu\text{m}$.

IL, ER, and crosstalk (CT) are also important parameters to measure the performance of the PSR. For the TM_0 and TE_0 modes, they are defined, respectively, as³⁵

$$\begin{aligned} IL_{\text{TM}} &= -10\lg(T_{\text{cro_TE}}), \\ IL_{\text{TE}} &= -10\lg(T_{\text{thr_TE}}), \end{aligned} \quad (3)$$

$$\begin{aligned} ER_{\text{TM}} &= 10\lg(T_{\text{cro_TE}}/T_{\text{cro_TM}}), \\ ER_{\text{TE}} &= 10\lg(T_{\text{thr_TE}}/T_{\text{thr_TM}}), \end{aligned} \quad (4)$$

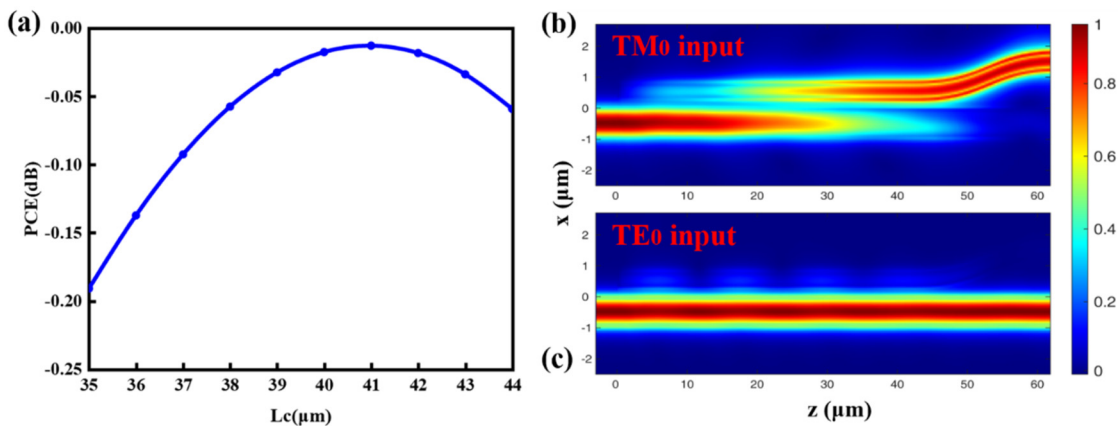


FIG. 3. (a) The variation of polarization conversion efficiency of the device with coupling length in the range of $35\text{--}44\text{ }\mu\text{m}$. Optical field transmission for (b) TM_0 mode input and (c) TE_0 mode input at determined waveguide widths $w_1 = 1\text{ }\mu\text{m}$, $w_2 = 0.55\text{ }\mu\text{m}$, and coupling length $L_c = 41\text{ }\mu\text{m}$.

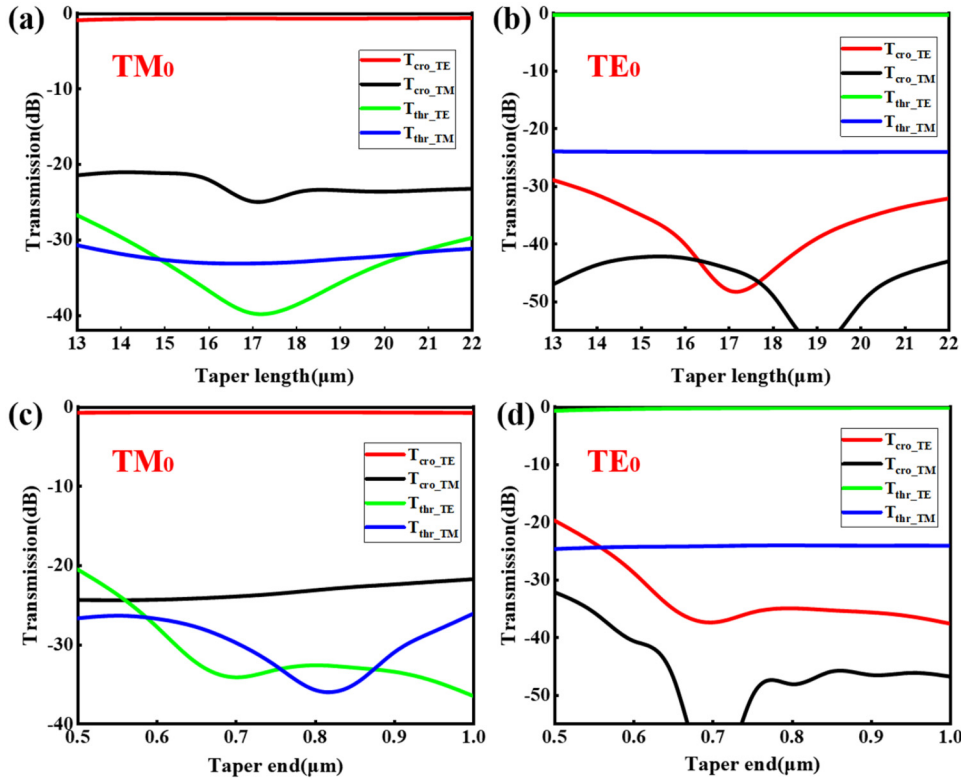


FIG. 4. Variation of transmissions with taper length in the range of 13–22 μm for (a) TM₀ mode and (b) TE₀ mode incidence. Transmissions at the taper end width in the range of 0.5–1 μm at an optimal taper length of 17 μm for (c) TM₀ mode and (d) TE₀ mode incidence.

$$\begin{aligned} CT_{TE} &= -10\lg(T_{thr_TE}/T_{cro_TE}), \\ CT_{TM} &= -10\lg(T_{cro_TE}/T_{thr_TE}), \end{aligned} \quad (5)$$

where the subscripts in IL, ER, and CT indicate when light is incident in the TM₀ or TE₀ mode.

Figure 5 demonstrates the wavelength dependence of conversion efficiency for TM₀ mode incidence and TE₀ mode incidence,

respectively. Figure 5(a) demonstrates the IL of 0.7 dB and the ER of 22.5 dB for the TM₀ mode at a wavelength of 1550 nm. In the range of 1500–1600 nm, the insertion losses of the TM₀ mode are less than 1.85 dB, and the ERs are larger than 22.5 dB, with the insertion losses in the range of 1550–1575 nm being less than 1 dB. For TE₀ mode incidence, the IL is 0.21 dB and the ER is 23.8 dB at a wavelength of 1550 nm. The insertion losses are less than 0.31 dB and the ERs are larger than 22.7 dB in the range of

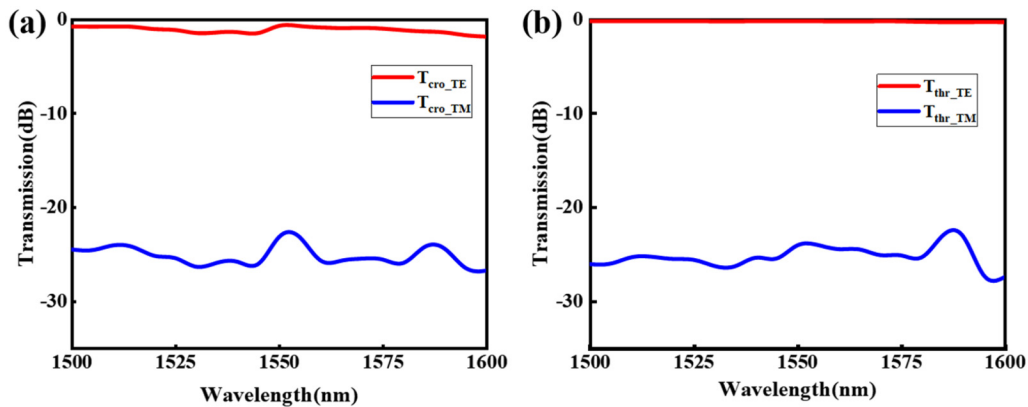


FIG. 5. Transmissions at wavelengths in the range of 1500–1600 nm for (a) TM₀ mode incidence and (b) TE₀ mode incidence.

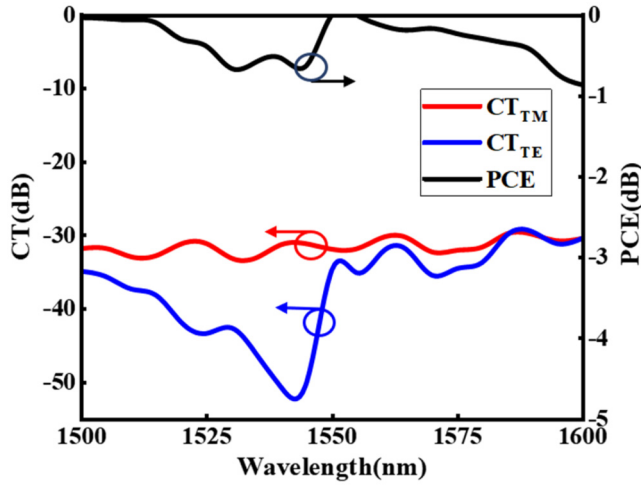


FIG. 6. The CTs (red line and blue line) and PCEs (black line) of the proposed PSR in the wavelength range of 1500–1600 nm.

1500–1600 nm, with the ERs larger than 25 dB in the range of 1500–1545 nm.

Figure 6 illustrates the CTs and PCEs of the PSR in the range of 1500–1600 nm. The CTs are less than -29.8 dB and -29.5 dB for the TM_0 mode and the TE_0 mode in the range of 1500–1600 nm, respectively. The PCE of the PSR is 99.97% at a wavelength of 1550 nm. Due to the different effective refractive indices of different wavelengths of light in the waveguide, there is a certain phase mismatch of light deviating from 1550 nm. The TM_0 mode cannot be sufficiently coupled to the adjacent waveguide to become the TE_0 mode, resulting in the existence of a partially unconverted TM_0 mode at the output port of WG1, and the PCE is affected. The PCEs are greater than 82% in the wavelength range of 1500–1600 nm, of which greater than 94% is maintained in about half of the wavelength range.

During the practical fabrication of the device, the sidewall roughness of the LN waveguide introduces scattered loss that affects the performance of the device.³⁶ The stress during processing affects both the surface and the interior of the material, and all these factors degrade the performance of the device.³⁷ In addition, during the actual fabrication process, the waveguide widths w_1 and

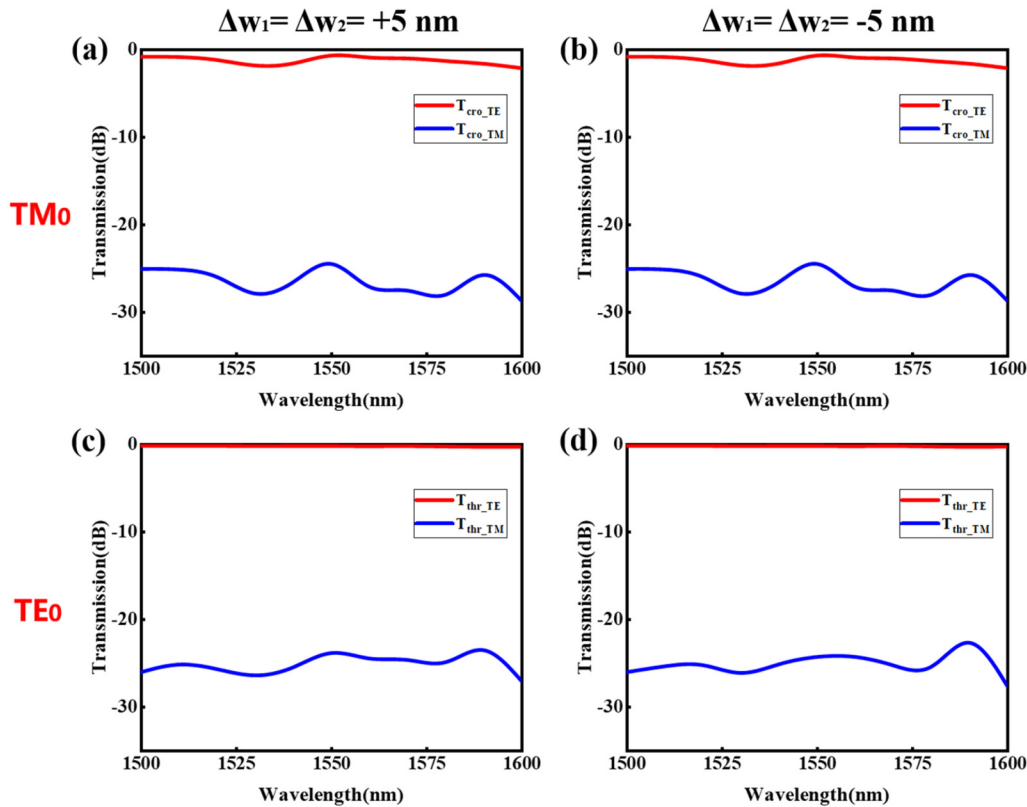


FIG. 7. Transmission spectra of (a) and (b) TM_0 mode and (c) and (d) TE_0 mode with w_1 and w_2 parameters simultaneously shifted by ± 5 nm.

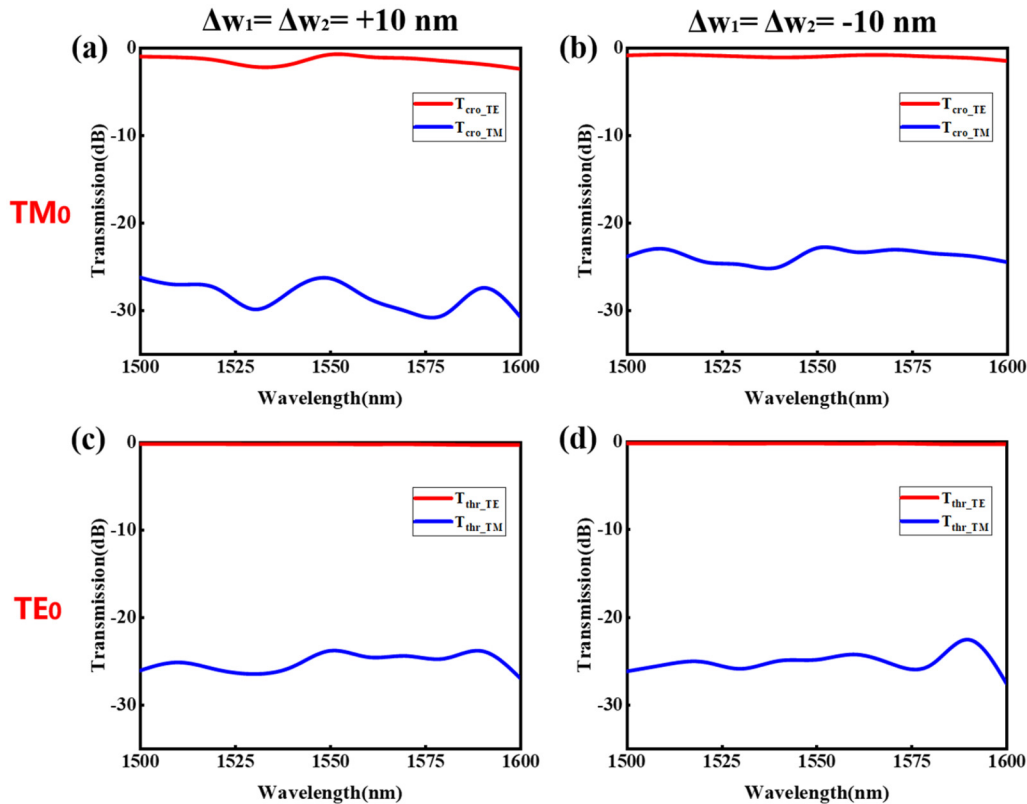


FIG. 8. Transmission spectra of (a) and (b) TM_0 mode and (c) and (d) TE_0 mode with w_1 and w_2 parameters simultaneously shifted by ± 10 nm.

w_2 in the coupling region are shifted simultaneously. The spacing g between the waveguides also changes, which may affect the performance of the PSR. Therefore, we choose $\Delta w_1 = \Delta w_2 = \pm 5$ nm and $\Delta w_1 = \Delta w_2 = \pm 10$ nm to quantify the fabrication tolerance of the PSR. Figures 7 and 8 show the transmission spectra of the TM_0 mode and the TE_0 mode simultaneously with ± 5 nm and ± 10 nm shifts in the waveguide widths w_1 and w_2 , respectively. For the TM_0 mode, the insertion losses in the 1500–1600 nm range are less than 2.2 dB, and the extinction ratios are higher than 22.0 dB when the parameters of w_1 and w_2 are varied at ± 5 nm. The insertion losses in the 1500–1600 nm range are less than 2.4 dB, and the extinction ratios are higher than 21.9 dB when the parameters of w_1 and w_2 are varied at ± 10 nm. Compared with Fig. 5(a) without a parameter shift, the mode coupling is reduced and the insertion loss becomes larger due to slightly shifted phase matching. For the TE_0 mode, the insertion losses in the wavelength range of 1500–1600 nm are less than 0.31 dB, and the extinction ratios are higher than 22.4 dB when the parameters of w_1 and w_2 are varied at ± 5 nm. The insertion losses in the wavelength range of 1500–1600 nm are less than 0.31 dB, and the extinction ratios are higher than 22.3 dB when the parameters of w_1 and w_2 are varied at ± 10 nm. Compared with Fig. 5(b), the insertion loss and extinction ratio of the device change slightly as the TE_0 mode does not undergo mode coupling,

and the slight change in the waveguide width does not affect the transmission of the optical field. In conclusion, the simulation results show that the simultaneous ± 5 nm and ± 10 nm variations of w_1 and w_2 have little effect on the device performance, which is beneficial for the fabrication of the devices.

Table I summarizes the simulation results of the designed PSRs based on the LNOI platform in recent years. Among all the PSRs, the proposed PSR has an acceptable extinction ratio in the

TABLE I. Simulation comparison of PSRs based on the LNOI platform.

Reference	ER (dB)	Bandwidth (nm)	IL (dB)	Length (μ m)
22	>26.8(TE_0) > 20(TM_0)	75	<0.2	440
23	>20	100	...	431
24	>18	230	<0.13	1240
25	>20(TE_0) > 10(TM_0)	111	<1.49	679
26	>26(TE_0) > 12(TM_0)	300	<1.7	271
27	>30	120	...	>1200
28	>18.7	50	...	240
This work	>22.7(TE_0) > 22.5(TM_0)	100	<1.85	61

100 nm bandwidth. Most importantly, the length of our proposed PSR is much shorter than that of the other PSRs, thanks to the approach of mode coupling that does not require a long adiabatic taper rather than the approach of mode evolution. The compact PSR structure facilitates its application in integrated photonic platforms.

III. CONCLUSION

In this work, a PSR cascaded with an ADC, a taper, and an S-bend is designed. The design of a mode-coupled PSR based on the LNOI platform is realized by simultaneously breaking the symmetry of the waveguide cross section using both the ridge waveguide and air cladding. The device features compactness and high ER with a length of only 61 μm . The device has an IL of less than 0.7 dB and an ER of larger than 22.5 dB at a wavelength of 1550 nm. In the range of 1500–1600 nm, the insertion losses are less than 1.85 dB and the ERs are larger than 22.5 dB for the TM_0 mode, and the insertion losses are less than 0.31 dB and the ERs are larger than 22.7 dB for the TE_0 mode. In addition, the CTs of this PSR in the 100 nm bandwidth are less than -29.5 dB, and the PCEs are greater than 82%. For the fabrication tolerance of this PSR, simulation results show that the performance of the device is only slightly affected when the variation of the waveguide widths w_1 and w_2 is ± 10 nm. The proposed compact PSR has great potential for applications in PDM transmitters and on-chip polarization controllers.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jiali Zhang: Software (lead); Visualization (equal); Writing – original draft (lead); Writing – review & editing (lead). **Kanglu Zhou:** Validation (equal). **Xiaochun Liu:** Visualization (equal). **Gui Chen:** Resources (equal). **Changshui Chen:** Project administration (lead); Supervision (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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